Replication of *The Efficacy of Seven Gratitude Interventions for Promoting Subjective Well-Being* by Dang et al. (2025, *Journal of Positive Psychology*)

Hew Yeh (mayeh@ucsd.edu)

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Introduction

Dang et al. (2025) sought to test 7 gratitude interventions that are used in positive psychology to improve subjective well-being. These interventions included intrapersonal, interpersonal, and gratitude-to-God interventions. While studies in the literature have examined these interventions before, such studies are heterogeneous in their selection of outcome measures for subjective well-being and only compare a few interventions against each other, making comparisons across excluded interventions difficult. By contrast, this study included a comprehensive battery of interventions and standardized outcome measures (i.e., state gratitude, state positive/negative

affect, and life satisfaction). As a result, this experiment allows for a direct comparison of efficacy between the gratitude interventions and can inform what gratitude practices are most helpful for the general public to incorporate into their lives.

In Study 1, Dang et al. (2025) found that writing a private gratitude letter was the most effective intervention in increasing positive affect. In this replication report, I wanted to replicate this finding and hoped to find a similarly large Hedges' g of around 0.41 when comparing this intervention to an active control. This replication study differs because we will test this effect in a student population as opposed to US adults from Prolific. This is aimed to increase the generalizability of the original study and offer guidance for students to improve their well-being. Furthermore, in case these interpersonal interventions caused increases to well-being because it reduces loneliness (rather than just improving gratitude), we also measure loneliness to include as a covariate to isolate the unique effect that gratitude has on improving well-being.

Methods

Sample Size/Power Calculation for Key Confirmatory Test

The original Dang et al. (2025) study compared post-test positive affect scores between a gratitude letter writing condition and an events list writing condition and obtained a Hedges' $g = 0.41$. This is the key effect we aim to replicate.

Hedges' g can be thought of as a bias-corrected form of Cohen's d because Cohen's d is upwardly biased (especially in small samples; Hedges, 2025; Liu, 2024). While the Dang et al. (2025) study reported the unbiased Hedges' g, many power analysis programs only offer sample size estimates based on Cohen's d. With larger samples, the difference between Hedges' g and Cohen's d becomes minimal as the $J(x)$ correction factor applied to Cohen's d approaches 1 with large n (Hedges, 1981, pp. 112–115)[1]. However, for the sake of precision, the Hedges' g was converted to a Cohen's d for the power/sample size analysis. The conversion is explained below:

Because g is the product of $J(x) \cdot d$, we can divide Hedges' g by the $J(x)$ correction factor to recover Cohen's d, where $x = n_1 + n_2 - 2$ for the correction factor $J(x)$. In Dang et al. (2025) Study 1, both conditions had exactly $n_1 = n_2 = 126$ participants, therefore $x = 126 + 126 - 2 = 250$.

[1] $J(x)$ is denoted as $c(m)$ in Hedges (1981).

$$
\begin{aligned}
J(x) \cdot d &= g \\
d &= \frac{g}{J(x)} \\
d &= \frac{g}{\frac{\Gamma(x/2)}{\sqrt{x/2} \times \Gamma((x-1)/2)}}
\end{aligned}
$$

$$
\begin{aligned}
&\#\text{Substitute } J(x) = J(250) \\
d &= g \times \frac{\sqrt{250/2} \times \Gamma((250-1)/2)}{\Gamma(250/2)} \\
d &= g \times \frac{\sqrt{125} \times \Gamma(124.5)}{\Gamma(125)} \\
d &= g \times 1.003013...
\end{aligned}
$$

```
# Calculate Correction Factor and Assign it to J_250
J_250 <- sqrt(125) * gamma(124.5) / gamma(125)

# Multiply g = 0.41 by the J(250) correction factor
ComputedCohens_d <- 0.41 * J_250
```

Now that the Cohen's d has been computed ($d = 0.411...$), we used the `pwr` package in R to perform a sample size analysis for 80% power.

```
# Step 1: Set CRAN as the source to install `pwr`
options(repos = c(CRAN = "https://cloud.r-project.org"))

# Step 2: Install `pwr`
install.packages("pwr")

# Step 3: Load `pwr` into the current session
library(pwr)

# Step 4: Calculate Power
Power80 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .80,
           type = "two.sample",
           alternative = "two.sided")
```

At an alpha level of $\alpha = .05$, for 80% power ($1 - \beta = .80$), we are going to need 94 participants per group. In total, that will be 188 participants.

For a balance of power and practicality, we will aim for a power level of at least 80% which will require 188 participants. An additional 10% margin will be collected in case of participant attrition for follow-up analyses which will result in collecting data until we reach 207 participants. Considering that the study is an online survey experiment that takes less than 30 minutes, this should balance both practicality and statistical validity power concerns.

Participants

The participant pool will be undergraduates from the University of California San Diego. There likely will be no adverse effects of the study. However, if any are reported by participants, data collection will be stopped immediately.

While we will not exclude human participants, we do not want to include bots or responses from large language models, and artificial intelligence (AI) agents, and as a result, invisible white text will be added into the prompt in an attempt to catch AI responses. If a participant responds to this prompt, their data will be row-wise deleted.

Materials & Procedure

Study materials include the jsPsych code (de Leeuw et al., 2023) used to run the experiment as well as the stimuli described below. Participants will also need access to a computer and an internet connection to complete the study.

After reading an informed consent form, consenting participants were welcomed into the study. They first filled out demographic information about their age, gender identity, and racial identity. Proceeding after that, participants answer a 1-item mood question and then are randomized to participate in either the events list or gratitude letter condition. Directly after, participants were asked to respond to questions that assessed post-intervention loneliness, gratitude, and positive affect. Lonliness was measured in one block with items randomized. Gratitude and positive affect were assessed simultaneously in an emotion matrix with their pooled items randomized like in the original study. Lastly, participants are thanked for their participation in the study and are debriefed on the study's aims and background literature.

In the interest of replication, I will use the same instructions that were provided in the original study (Cunha et al., 2019; Dang et al., 2025; Emmons & McCullough, 2003; Walsh et al., 2022). After participants are randomly assigned to a condition, they will see the following directions:

- **Experimental Condition: Gratitude Letter (Walsh et al., 2022):** "In our daily lives, we often thank people for both large and small things. However, it can sometimes be difficult to make time to express our gratitude more deeply. Today, we want you to write a letter of gratitude to someone who has done something for which you are extremely grateful. Please do not share your letter with this person or anyone else."

- **Control Group: Events List (Cunha et al. 2019):** "In life, there are events, both large and small, that end up affecting us. Think back over the past day and write down five events that somehow affected you."

Participants will also be told the following instructions from the original study (Dang et al., 2025):

> As you write, don't worry about perfect grammar and spelling, and remember that anything you write will remain strictly confidential. Should an experimenter read this entry in the future, it will be identifiable only by a participant number and not by your name.

After participants finish the intervention, they will be asked to fill out a questionnaire that asks them about their state gratitude, state loneliness, and state positive affect. The order of items within each block are randomized. Next, participants are asked to pass a prompt injection AI captcha. Lastly, participants will be debriefed about the study goals and be given on-campus mental health resources. Finally, participants were thanked for their participation and awarded research credits[2] that can be used to fulfill course research participation requirements or confer extra credit.

Measures

Demographics

Participants were asked to indicate their age, gender identity, and racial identity.

Mood

At pre-test, participants were asked a 1-item question "How are you feeling right now?" with responses ranging from (1) *Very Bad* to (7) *Very Good*.

State Loneliness

On a scale from (1) *Not at All* to (7) *Very Much*, participants indicated their agreement with the following questions: "How lonely are you feeling? Compared to other people, how lonely do you think you are? To what extent do you feel lonely right now?" Question order was randomized. This state measure of loneliness was originally used by Choi et al. (2016) as an adapted version of a measure used by Shaver et al. (1985) and demonstrated good internal consistency in the Choi et al. (2016) study (Cronbach's $\alpha = .92$). It was important that this measure of loneliness was a *state* measure as opposed to a *trait* measure as we wanted to see if momentary fluctuations in loneliness as a result of our intervention may explain some of the effect that our writing conditions had on positive affect.

[2] In this case, it will be 0.25 SONA credits for a roughly 30-minute study.

State Gratitude

On a scale from (1) *Strongly Disagree* to (7) *Strongly Agree*, participants rated how much they agreed with three adjectives: "grateful," "appreciative," and "thankful" (Dang et al., 2025). The order of these items were randomized. This scale also demonstrated good internal consistency in the original study (McDonald's $\omega = .93$).[3]

State Positive Affect

On a scale from (1) *Strongly Disagree* to (7) *Strongly Agree*, participants rated how much they agreed with five adjectives: "happy," "joyful," "pleased," "content," and "satisfied" (Dang et al., 2025) with the order of the adjectives randomized. The original study demonstrated high internal consistency for this measure (McDonald's $\omega = .95$). Positive affect was chosen as one dimension of subjective well-being to measure; however, theoretically speaking, negative affect and overall life satisfaction are other key components to subjective well-being (Diener et al., 2022).

Analysis Plan

Statistical Software and Analyses

Data analyses will be performed using the R programming language (R Core Team, 2026). Our key confirmatory test will be a t-test comparing average positive affect levels across the letter writing (experimental) vs. event writing (control) conditions. Using a alpha threshold of $\alpha = .05$ that was chosen during our power/sample size analysis, I will use the `t.test` function to compute the t-test test statistic. This function will also ouput a Cohen's d which I will convert to Hedges' g using Hedges' $J(x)$ correction where x is the degrees of freedom for the t-test. This will represent the intent-to-treat effect that most closely matches Dang et al. (2025).

For the exploratory analysis that excludes participants who spend a total time of less than 2 standard deviations away from the median in the study, we will perform the same t-test procedure but filter the data first to exclude participants that do not meet the 2 standard deviation criteria.

For the exploratory analyses involving structural equation modeling, I will use lavaan (version 0.7-3; 2026, October 8th release), a popular and free R package to estimate multivariate models (e.g., path analysis) and perform latent variable analysis (e.g., confirmatory factor analysis, structural regression, and latent growth models; Rosseel, 2012).

[3] McDonald's Omega describes the proportion of variance in the indicators that is explained by a general latent factor compared to the overall amount of variance in the indicators with the highest score being 1. $\omega = \frac{\left(\sum_{i=1}^{k} \lambda_i\right)^2}{\left(\sum_{i=1}^{k} \lambda_i\right)^2 + \sum_{i=1}^{k} \psi_i}$, where $\psi := 1 - \lambda_i^2$.

Data and code will be made publicly available. However, since the events and letters are quite personal and we promised participants confidentiality, we will remove their responses to free-response questions (i.e., the written response to the event or letter writing condition). The only remaining information will be a newly randomly generated participant ID, their age, gender, ethnoracial identity, their scores across the likert scales, and how long they took to complete the study. Additionally, free-response answers to gender identity or ethnoracial identity will be recoded as "Other" unless at least 10 participants endorse a given response.

The original study used an intent-to-treat analysis which meant that they did not exclude any participants from the study. This is aimed to measure the generalizable real-world impact of an intervention and acknowledge that attrition and non-compliance are inseparable downstream effects of the intervention. Since this study is meant to inform real student populations, it is vital to understand the realistic intent-to-treat effect size as opposed to the effect size found in an artificially controlled setting. Thus, I opted to also use an intent-to-treat analysis technique.

Differences from Original Study

Key differences from the original study mainly include using SONA for data collection among UC San Diego undergraduates and only having 2 conditions as opposed to the 11 in the original study.

Results

Data Preparation

Aligned with the Dang et al. (2025) paper analysis protocol, data will be analyzed without excluding participants to measure the intent-to-treat effect. Prior to running analyses, responses will be scanned for AI flags (viz., the gamma(125) AI catch). Participants that fail the AI check will be excluded entirely and be row-wise deleted. Individual responses will also be checked prior to dropping to ensure that responses did not actually contain that number for legitimate reasons.

```
library(dplyr)
Tensor <- read_csv("")

# Look at the rows that have these in there to drop
Tensor <- Tensor[grepl("1.5", Tensor$COL), ]
Tensor <- Tensor[grepl("15", Tensor$COL), ]

# Drop the appropriate participant right here.
```

```r
gamma(125)
```

```
[1] 1.506142e+207
```

Confirmatory Analysis

```r
MainEffect <- t.test(DV ~ Condition, data = Tensor)

# Calulate Hedges' g using J(x) and Cohen's d
```

A t-test was computed to assess whether the average positive affect score differed between the gratitude letter and events list conditions, $t(df) = XX.XX$, $p < or = .XX$, Hedges' $g = GG.GG$. There was/was not a significant difference such that the _____ group had higher mean positive affect scores than the _____ group.

Exploratory Analyses

Exploratory Analysis 1: Excluding Unreasonably Fast Responses

While the intent-to-treat analyis shows a real-world repsonse to the intervention, since we have the data, we can also calculate the effect while excluding people who may have sped past in the study. This effect size will be interpretted as the effect of the intervention under ideal conditions. For this analysis, participants who spent a total time in the study of less than 2 standard deviations under the median will be exlucded. This exclusion criteria is used commonly in survey research and is the criteria that Qualtrics typically uses to identify participants who completed the task abnormally fast (often called "speeders"; Qualtrics, 2026). This is to test whether the observed intent-to-treat effect is attenuated by participants who speed past the intervention and questions without giving it much thought.

Exploratory Analysis 2: Putative Mediation

aThe original study demonstrates a dosage effect where interventions which brough about greater amounts of gratitude also led to higher levels of positive affect. This raises the question of whether increasing gratitutde (the goal of the intervention) is the mechanism that leads to increases in positive affect. As a result, I will construct a mediation model that tests whether

gratitude may be a putative mediator and if this explains the effect beyond making someone feel more connected when writing a letter by also adding loneliness as another putative mediator[4].

We will run a structural equation model predicting latent positive affect scores using experimental condition while specifying latent state gratitude and latent state loneliness scores as putative mediators. By using latent factor scores, we also can avoid assuming that all items in each of the scales are equally informative and allow the natural covariance structure of the data to determine how heavily to weigh each item. Since condition is a categorical indicator, we will only standardize the coefficients with respect to the latent factor scores. Thus, each effect size represents the expected difference in latent factor scores in standardized units between the conditions.

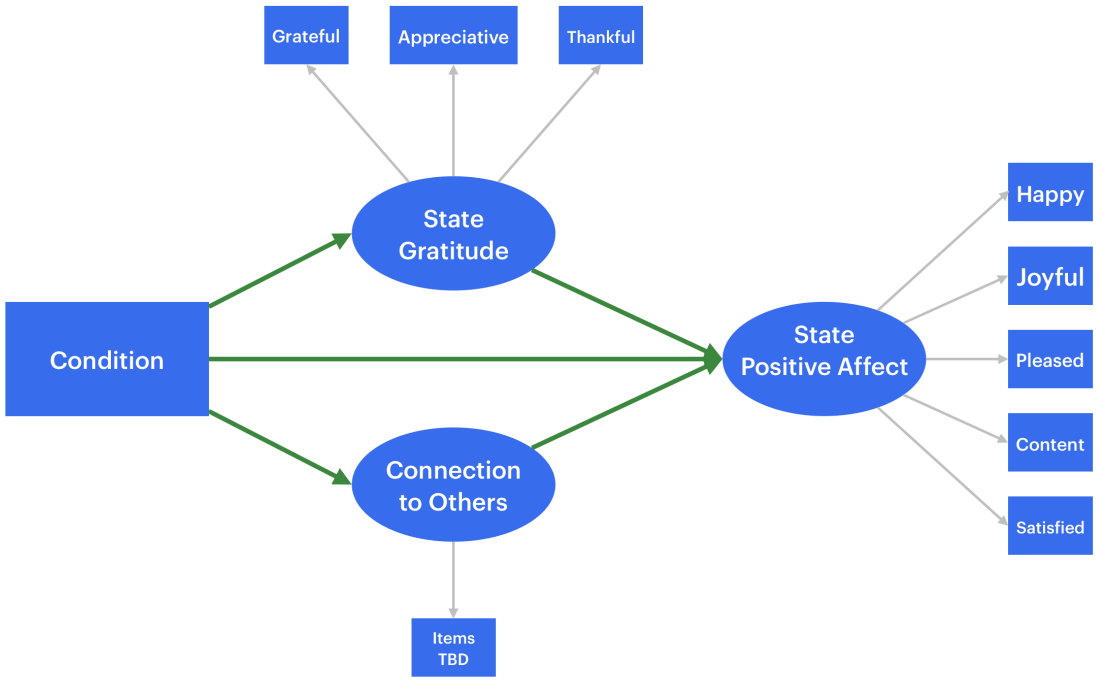


Figure 1: Proposed Structural Equation Model With Dual Putative Mediators. Measurement Paths are in Gray. Structural Paths are in Green.

`Syntax for SEM`

[4]This strategy suffers from some issues in causal inference especially since loneliness was measured after positive affect as we were worried that the items used to measure loneliness would negatively affect people's level of positive affect. In the spirit of replication, we also needed to measure gratitude and positive affect directly after the intervention to match the original study. Future studies should attempt to rule out alternative casusal possibilities such as increases in positive affect leading to greater gratitude and a reduced sense of loneliness.

```
# Measurement
PA          BY PA1 PA2 PA3 PA4 PA5
GRAT        BY GRAT1 GRAT2 GRAT3
CONNECTION  BY ITEMS

# Structural Regression
PA          ON GRAT COND CONNECTION
GRAT        ON COND
CONNECTION  ON COND
```

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